



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
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Roll Number	2520030142
Branch	CSE
Course Outcome	CO-2 (B+ Tree)

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using B+ Tree

Work Done Summary:

Waste Sense – Smart Waste Collection & Recycling Optimization System Using B+ Tree

The Waste Sense system uses a **B+ Tree** to efficiently manage and index smart waste-bin records in the database.

Work Done:

- Developed a B+ Tree-based indexing system for storing smart waste-bin information.
- Organized waste-bin records using unique Bin IDs as indexing keys.
- Implemented efficient insertion of new smart-bin records into the B+ Tree.

Implementation Details:

- Each city zone stores daily waste collection data.
- A Segment Tree is built using waste quantities.
- Range queries calculate total waste generated between selected zones.
- Updates are performed whenever new sensor readings arrive.
- The system supports real-time monitoring of waste accumulation.

Result: Screenshot/output:

```
1. Insert Bin Record
2. Search Bin Record
3. Delete Bin Record
4. Display All Bin Records
5. Range Query
6. Exit

Enter Choice: 1
Enter Bin ID: 101
Record Inserted Successfully.

Enter Choice: 1
Enter Bin ID: 105
Record Inserted Successfully.

Enter Choice: 1
```

- The Waste Sense system successfully stores and manages smart waste-bin records using a **B+ Tree**. Bin IDs are indexed efficiently, allowing fast insertion, searching, updating, and deletion of waste-bin records. Range queries are performed effectively to retrieve bins within specific zones or fill-level ranges.

GIT HUB LINK : <https://github.com/nemaliscreddy78-glitch/DSA2>

Student Signature	Faculty Signature